Supplementary Evidence

This supplement reports existing fitted-model results. No new model fits were performed for these tables. Synthetic intervals use paired replications; benchmark comparisons use dataset means, not folds as independent observations.

S1. Paired intervals for the targeted extension

Differences are Complete minus comparator. Each setting has 30 paired replications. Percentile intervals use 5000 paired bootstrap resamples. They are exploratory individual 95% intervals, with no simultaneous-coverage or multiplicity-adjusted significance claim. Full default-rule, ranking and additional-comparator intervals are included in the accompanying CSV.

Table S1: Selected-threshold G-mean differences: LR

Geometry / noise	Comparator	Mean difference	Individual 95% CI
Gaussian / clean	SMOTE	-0.00001	$[-0.00184, +0.00190]$
Gaussian / clean	Matched SMOTE	-0.00061	$[-0.00272, +0.00143]$
Gaussian / dispersed	SMOTE	+0.00016	$[-0.00233, +0.00260]$
Gaussian / dispersed	Matched SMOTE	-0.00002	$[-0.00254, +0.00236]$
Gaussian / boundary	SMOTE	+0.00002	$[-0.00163, +0.00176]$
Gaussian / boundary	Matched SMOTE	-0.00015	$[-0.00160, +0.00135]$
Moons / clean	SMOTE	-0.01222	$[-0.01685, -0.00756]$
Moons / clean	Matched SMOTE	-0.01247	$[-0.01703, -0.00789]$
Moons / dispersed	SMOTE	+0.00557	$[+0.00190, +0.00987]$
Moons / dispersed	Matched SMOTE	+0.00576	$[+0.00223, +0.00988]$
Moons / boundary	SMOTE	-0.01194	$[-0.02295, -0.00397]$
Moons / boundary	Matched SMOTE	-0.01466	$[-0.02570, -0.00635]$
Circles / clean	SMOTE	-0.00086	$[-0.00397, +0.00228]$
Circles / clean	Matched SMOTE	-0.00139	$[-0.00466, +0.00179]$
Circles / dispersed	SMOTE	-0.00229	$[-0.00579, +0.00090]$
Circles / dispersed	Matched SMOTE	-0.00122	$[-0.00449, +0.00194]$
Circles / boundary	SMOTE	-0.00020	$[-0.00168, +0.00116]$
Circles / boundary	Matched SMOTE	-0.00047	$[-0.00178, +0.00073]$

Table S2: Selected-threshold G-mean differences: SVM

Geometry / noise	Comparator	Mean difference	Individual 95% CI
Gaussian / clean	SMOTE	-0.00516	$[-0.00971, -0.00082]$
Gaussian / clean	Matched SMOTE	-0.00658	$[-0.01101, -0.00222]$
Gaussian / dispersed	SMOTE	+0.00439	$[-0.00122, +0.01057]$
Gaussian / dispersed	Matched SMOTE	-0.00245	$[-0.00596, +0.00104]$
Gaussian / boundary	SMOTE	+0.00062	$[-0.00323, +0.00506]$
Gaussian / boundary	Matched SMOTE	-0.00690	$[-0.01045, -0.00343]$
Moons / clean	SMOTE	-0.00121	$[-0.00215, -0.00031]$
Moons / clean	Matched SMOTE	-0.00100	$[-0.00193, -0.00011]$
Moons / dispersed	SMOTE	+0.00550	$[+0.00307, +0.00805]$
Moons / dispersed	Matched SMOTE	+0.00355	$[+0.00182, +0.00537]$
Moons / boundary	SMOTE	-0.01869	$[-0.02181, -0.01559]$
Moons / boundary	Matched SMOTE	-0.02007	$[-0.02333, -0.01687]$
Circles / clean	SMOTE	-0.00129	$[-0.00226, -0.00031]$
Circles / clean	Matched SMOTE	-0.00119	$[-0.00212, -0.00027]$
Circles / dispersed	SMOTE	+0.00502	$[+0.00071, +0.01085]$
Circles / dispersed	Matched SMOTE	+0.00000	$[-0.00109, +0.00115]$
Circles / boundary	SMOTE	-0.00330	$[-0.00609, -0.00107]$
Circles / boundary	Matched SMOTE	-0.00334	$[-0.00592, -0.00121]$

S2. All benchmark baselines

Only datasets with all 15 outer folds completed for a method enter its mean. D counts such datasets. Different D values indicate different populations; these method-wise averages must not be interpreted as a common-dataset ranking. WRND denotes the documented author-code adapter.

Table S3: All methods: LR, supplied labels

Method	D	G-mean	ROC-AUC	AP
No intervention	6	0.8760	0.9551	0.8851
Random oversampling	6	0.9150	0.9558	0.8879
SMOTE	6	0.9164	0.9542	0.8890
Borderline-SMOTE1	6	0.9136	0.9531	0.8781
ADASYN	5	0.8970	0.9463	0.8616
Complete	6	0.8953	0.9544	0.8908
No downweighting	6	0.8920	0.9544	0.8889
WRND adapter	3	0.9287	0.9583	0.8424

Table S4: All methods: RF, supplied labels

Method	D	G-mean	ROC-AUC	AP
No intervention	6	0.9029	0.9791	0.9350
Random oversampling	6	0.9279	0.9774	0.9204
SMOTE	6	0.9303	0.9790	0.9233
Borderline-SMOTE1	6	0.9215	0.9783	0.9210
ADASYN	5	0.9170	0.9712	0.8992
Complete	6	0.9228	0.9795	0.9262
No downweighting	6	0.9153	0.9794	0.9296
WRND adapter	3	0.8715	0.9711	0.8740

Table S5: All methods: SVM, supplied labels

Method	D	G-mean	ROC-AUC	AP
No intervention	6	0.9053	0.9760	0.9231
Random oversampling	6	0.9390	0.9800	0.9262
SMOTE	6	0.9342	0.9799	0.9276
Borderline-SMOTE1	6	0.9292	0.9780	0.9168
ADASYN	5	0.9296	0.9752	0.9048
Complete	6	0.9252	0.9787	0.9228
No downweighting	6	0.9276	0.9793	0.9232
WRND adapter	3	0.9034	0.9750	0.8729

Table S6: All methods: LR, 10 percent fitting-label corruption

Method	D	G-mean	ROC-AUC	AP
No intervention	6	0.8467	0.9466	0.8740
Random oversampling	6	0.8980	0.9499	0.8712
SMOTE	6	0.9012	0.9487	0.8772
Borderline-SMOTE1	6	0.8715	0.9432	0.8556
ADASYN	6	0.8848	0.9481	0.8667
Complete	6	0.9081	0.9509	0.8830
No downweighting	6	0.9090	0.9526	0.8868
WRND adapter	3	0.9188	0.9547	0.8263

Table S7: All methods: RF, 10 percent fitting-label corruption

Method	D	G-mean	ROC-AUC	AP
No intervention	6	0.8971	0.9667	0.9155
Random oversampling	6	0.9046	0.9618	0.8989
SMOTE	6	0.9214	0.9653	0.9075
Borderline-SMOTE1	6	0.9040	0.9580	0.8855
ADASYN	6	0.9100	0.9587	0.8891
Complete	6	0.9233	0.9729	0.9192
No downweighting	6	0.9170	0.9724	0.9206
WRND adapter	3	0.8861	0.9577	0.8592

Table S8: All methods: SVM, 10 percent fitting-label corruption

Method	D	G-mean	ROC-AUC	AP
No intervention	6	0.9010	0.9655	0.9135
Random oversampling	6	0.9316	0.9760	0.9105
SMOTE	6	0.9320	0.9766	0.9131
Borderline-SMOTE1	6	0.8975	0.9539	0.8632
ADASYN	6	0.9192	0.9667	0.8922
Complete	6	0.9259	0.9767	0.9145
No downweighting	6	0.9271	0.9794	0.9216
WRND adapter	3	0.9074	0.9711	0.8619

S3. Comparisons on common complete datasets

Each row restricts Complete and its comparator to the same datasets, with 15 complete outer folds for both. Differences are Complete minus comparator. No new significance test is introduced by these descriptive tables.

Table S9: Paired means on common datasets: LR, supplied labels

Comparator	D	Δ G-mean	Δ ROC-AUC	Δ AP
ADASYN	5	-0.02025	-0.00102	+0.00745
Borderline-SMOTE1	6	-0.01826	+0.00129	+0.01270
No downweighting	6	+0.00327	-0.00001	+0.00199
No intervention	6	+0.01934	-0.00073	+0.00578
Random oversampling	6	-0.01969	-0.00140	+0.00292
SMOTE	6	-0.02107	+0.00021	+0.00189
WRND adapter	3	-0.04762	+0.00239	+0.00235

Table S10: Paired means on common datasets: RF, supplied labels

Comparator	D	Δ G-mean	Δ ROC-AUC	Δ AP
ADASYN	5	-0.00882	+0.00423	+0.01227
Borderline-SMOTE1	6	+0.00124	+0.00117	+0.00517
No downweighting	6	+0.00749	+0.00005	-0.00344
No intervention	6	+0.01989	+0.00035	-0.00884
Random oversampling	6	-0.00517	+0.00210	+0.00572
SMOTE	6	-0.00757	+0.00043	+0.00282
WRND adapter	3	+0.02118	+0.00042	-0.00460

Table S11: Paired means on common datasets: SVM, supplied labels

Comparator	D	Δ G-mean	Δ ROC-AUC	Δ AP
ADASYN	5	-0.01940	-0.00076	+0.00259
Borderline-SMOTE1	6	-0.00409	+0.00074	+0.00604
No downweighting	6	-0.00241	-0.00058	-0.00038
No intervention	6	+0.01990	+0.00269	-0.00024
Random oversampling	6	-0.01389	-0.00130	-0.00336
SMOTE	6	-0.00904	-0.00117	-0.00480
WRND adapter	3	-0.00228	-0.00171	-0.00501

Table S12: Paired means on common datasets: LR, corrupted fitting labels

Comparator	D	Δ G-mean	Δ ROC-AUC	Δ AP
ADASYN	6	+0.02334	+0.00283	+0.01625
Borderline-SMOTE1	6	+0.03663	+0.00773	+0.02735
No downweighting	6	-0.00089	-0.00175	-0.00377
No intervention	6	+0.06145	+0.00433	+0.00894
Random oversampling	6	+0.01009	+0.00096	+0.01182
SMOTE	6	+0.00697	+0.00215	+0.00577
WRND adapter	3	-0.00242	+0.00135	+0.00703

Table S13: Paired means on common datasets: RF, corrupted fitting labels

Comparator	D	Δ G-mean	Δ ROC-AUC	Δ AP
ADASYN	6	+0.01331	+0.01421	+0.03005
Borderline-SMOTE1	6	+0.01925	+0.01490	+0.03365
No downweighting	6	+0.00631	+0.00051	-0.00142
No intervention	6	+0.02623	+0.00614	+0.00364
Random oversampling	6	+0.01867	+0.01106	+0.02030
SMOTE	6	+0.00195	+0.00760	+0.01171
WRND adapter	3	+0.01401	+0.00448	+0.00165

Table S14: Paired means on common datasets: SVM, corrupted fitting labels

Comparator	D	Δ G-mean	Δ ROC-AUC	Δ AP
ADASYN	6	+0.00668	+0.00999	+0.02230
Borderline-SMOTE1	6	+0.02847	+0.02277	+0.05136
No downweighting	6	-0.00113	-0.00270	-0.00701
No intervention	6	+0.02491	+0.01124	+0.00106
Random oversampling	6	-0.00563	+0.00074	+0.00407
SMOTE	6	-0.00606	+0.00005	+0.00140
WRND adapter	3	-0.00265	-0.00027	-0.00677

S4. Completed and failed benchmark entries

Counts refer to classifier–outer-fold–condition entries, not necessarily distinct resampling events. Failures are retained and are not assigned scores.

Table S15: Benchmark Completion and Failure Counts

Method	Successful	Failed
ADASYN	501	39
Borderline-SMOTE1	540	0
No downweighting	540	0
No intervention	540	0
Complete	540	0
Random oversampling	540	0
SMOTE	540	0
WRND adapter	366	174